Syllabus

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

ECE 410/510
Spring 2026

Course information:

Class sessions:	Mon/Wed, 03:30pm – 05:10pm
Term:	Mar 30 – Jun 7, 2026
Holidays:	Mon, May 25, 2026 (Memorial Day)
Finals week:	Jun 8 – 13, 2026
Credits	4
Prerequisites:	Undergraduate: ECE 371 and ECE 351 Graduate: ECE 485

Your instructor:

Name:	Christof Teuscher, PhD
E-mail:	teuscher@pdx.edu I always try to reply to student e-mails promptly, but please do not assume that you will get a reply within hours, especially not after 5pm and on weekends. Due to the volume of e-mails I get, my response times can be longer than 24h.
Office:	FAB 160-07 4th Avenue Building (1900 SW 4th Avenue) Suite 160 (ECE main offices)
Office hours:	Pick a meeting time at http://www.teuscher-lab.com/officehour

Course description:

This course covers the design, simulation, optimization, and evaluation of specialized hardware for artificial intelligence and machine learning workloads. Students begin with the mathematical foundations that connect AI/ML algorithms to their hardware implementations, then apply HW/SW co-design methods to map algorithms such as CNNs, DNNs, and transformer-based LLMs onto architectures including GPUs, TPUs, FPGAs, systolic arrays, and neuromorphic processors. Additional topics include computational profiling with Python and CUDA, benchmarking across hardware platforms, in-memory computing with memristive crossbar arrays, and the use of LLMs as hardware design tools for HDL generation and physical design automation. The course alternates between lectures and hands-on codefest sessions where students tackle open-ended design challenges, from implementing neurons and systolic sorting to running simulations on neuromorphic hardware platforms such as BrainScaleS-2.

Course organization:

The course follows a weekly cycle: Monday lectures introduce concepts, methods, and context, while Wednesday codefest sessions are hands-on working sessions where students tackle structured design challenges that apply and extend the lecture material. Each codefest contains two sequential challenges: CMAN must be completed without AI assistance and verifies foundational understanding; CLLM scales the same concept to real-world complexity where AI tooling is required. The main project is individual work. Students are expected to use LLMs and other modern tools during codefests and project work.

Why the syllabus is important and other important things:

- **This syllabus contains information that will be crucial to your success in this course. Read this whole syllabus carefully, keep it accessible, and refer to it regularly throughout the course.**

- **The syllabus is your contract with your instructor. If you enroll in this course, you agreed to the rules and regulations as outlined in this syllabus.**
- The syllabus may be adjusted during the term (see flexibility statement below). Check Canvas for changes. Canvas always contains the most recent and up-to-date information about this course.
- **You are expected to read your pdx.edu e-mail daily.**

Flexibility statement:

The instructor reserves the right to modify course content and/or substitute assignments and learning activities in response to institutional, weather, or class situations.

Course websites and learning platforms:

- We will be using Canvas, PSU's official Learning Management System (LMS). Canvas can be accessed at the following URL: https://canvas.pdx.edu.
- You are expected to know all parts of the course material posted on Canvas.
- **You are expected to check Canvas for updates and news daily.**
- Canvas comes with a great notification system that you can customize. Set it up according to your preferences so that you never miss a deadline, message, etc.

Required materials:

- There is **no** required book for this term.
- You need to be able to bring a **laptop** to class for the Wednesday codefests.
- This course is built around LLM-assisted hardware design ("vibe coding"). You will use AI tools to design, simulate, and synthesize custom AI accelerator hardware in a hardware description language, even if you have limited prior experience with hardware description languages. **This means your AI tools are not a nice-to-have; they are essential to how the course works**. To get the most out of this course, you need a paid subscription to at least one frontier LLM. The free tiers impose rate limits and model restrictions that will get in your way during codefests and project work. I recommend one of the following:
    - Claude Pro ($20/mo, or ~$17/mo billed annually): https://claude.ai/pricing
    - Google Gemini Advanced / Google AI Pro ($19.99/mo): https://one.google.com/explore-plan/gemini-advanced
    - ChatGPT Plus ($20/mo): https://chatgpt.com
- All three are capable of generating Python, Verilog, SystemVerilog, writing testbenches, and assisting with ASIC design workflows. You are welcome to use more than one. I personally find Claude the strongest for code generation and reasoning about hardware architectures, but you should experiment and find what works best for your workflow.
- Student discounts and free access:
    - Google Gemini Advanced: only a free 1-month trial is offered for our region currently: https://gemini.google/students
    - ChatGPT: OpenAI does not currently offer a standing student discount, but the free tier and the Go plan ($8/mo) are available.
    - Claude: Anthropic does not offer a formal student discount at this time. Claude Pro is $20/mo ($17/mo annual).
- Note that Github copilot has recently downgraded the student plan. Some premium models, including GPT-5.4, and Claude Opus and Sonnet, are no longer available under the student plan. Relying on Github copilot for this course is therefore not recommended.

Learning outcomes:

- Apply the mathematical foundations underlying AI/ML algorithms (linear algebra, calculus, probability) in the context of hardware design and optimization.

- Explain the principles of HW/SW co-design and justify partitioning decisions for AI/ML workloads.
- Map AI/ML algorithms onto specialized hardware through co-design methodologies.
- Evaluate and optimize HW designs through co-design for computational power, energy efficiency, and flexibility.
- Explain the foundations of neural networks and Large Language Models (LLMs), including CNNs, DNNs, and transformer architectures.
- Describe the architectures of specialized AI/ML hardware including GPUs, TPUs, FPGAs, neuromorphic processors, and in-memory computing systems.
- Design, implement, and verify a custom hardware accelerator for an AI/ML algorithm using a hardware description language (Verilog or SystemVerilog).
- Use modern software and hardware tools, including LLM-assisted design flows, for designing and deploying specialized AI/ML hardware.
- Critically evaluate LLM-generated hardware descriptions for correctness, efficiency, and synthesizability.
- Use the ASIC design flow (RTL to GDSII) including synthesis, placement, routing, and timing analysis.
- Benchmark hardware implementations against software baselines using appropriate performance metrics.
- Explain emerging computing paradigms, including in-memory computing, memristive devices, and their applications to AI/ML workloads.
- Communicate design decisions, tradeoffs, and results through professional documentation.

Course schedule (tentative):

The actual course schedule is available on Canvas. A summary is included below. Please note that things can and will change. Any updates and changes to the course schedule will be reflected on Canvas.

Wk	Date	Day	Type	Topic / activity	Project / milestone
1	Mar 30	Mon	Lecture	Introduction: What is computation? What is a computer? Course organization. AI/ML landscape overview.	Identify candidate algorithms; review application list.
1	Apr 1	Wed	Codefest	CMAN (no AI): MAC/parameter/activation memory count by hand for a 3-layer MLP. CLLM (AI OK): Environment setup (GitHub, Slack). Profile ResNet-18 with torchinfo. Draft Heilmeier answers. Select project topic.	Draft Heilmeier answers; identify algorithm; run initial profiler; confirm SW baseline. Topic selected.
2	Apr 6	Mon	Lecture	HW/SW co-design fundamentals: partitioning, profiling bottlenecks, interface design, tradeoff analysis. Roofline model: arithmetic intensity, compute-bound vs. memory-bound regimes, ridge point.	Revisit HW/SW partition with roofline framing.
2	Apr 8	Wed	Codefest	CMAN (no AI): Roofline construction by hand; classify GEMM and vector-add kernels. CLLM (AI OK): CNN inference profiling; plot kernels on roofline; HW/SW partition proposal.	**M1 DUE Sun, Apr 12**: Heilmeier answers, profiling, roofline, HW/SW partition, interface selection, SW benchmark.
3	Apr 13	Mon	Lecture	GPU architecture: SIMT execution, warp scheduling, memory hierarchy, CUDA programming model. Memory wall: DRAM vs. SRAM, bandwidth constraints, energy cost of data movement.	Identify which operations in your algorithm map to SIMT or tensor-core patterns.
3	Apr 16	Wed	Codefest	CMAN (no AI): DRAM traffic analysis for naive vs. tiled 64×64 matrix multiply. CLLM (AI OK): CUDA GEMM naive vs. tiled; Nsight profiling; roofline placement.	Draft HW architecture due: block diagram, data path sketch, first HDL scaffold.

Wk	Date	Day	Type	Topic / activity	Project / milestone
4	Apr 20	Mon	Lecture	Numerical representation and precision: FP32, FP16, BF16, INT8; quantization mechanics; multiplier area scales quadratically with bit width. DNNs on GPUs: CNN pipeline, backpropagation, tensor cores, mixed-precision training.	Decide on precision format and quantization strategy for target workload.
4	Apr 22	Wed	Codefest	CMAN (no AI): Manual INT8 symmetric quantization of a 4×4 FP32 weight matrix. CLLM (AI OK): LLM code review — evaluate and fix LLM-generated INT8 MAC unit in Verilog.	Build primary compute unit in HDL; write first testbench stub.
5	Apr 27	Mon	Lecture	TPUs, transformers, and systolic arrays: dataflow patterns, weight/output/input-stationary, self-attention mechanism.	Map your architecture to a systolic or dataflow pattern from lecture.
5	Apr 29	Wed	Codefest	CMAN (no AI): Cycle-by-cycle trace of a 2×2 systolic array. CLLM (AI OK): First OpenLane 2 synthesis attempt on project compute core; extract timing/area metrics.	**M2 DUE Sun, May 3**: First working simulation with testbench; interface module with testbench.
5			Quiz	Quiz 1: Canvas-based, open-note, individual. Covers lecture material through Week 4.	
6	May 4	Mon	Lecture	In-memory computing and spiking neural networks: crossbar arrays, LIF neurons, STDP, event-driven computation.	Evaluate whether in-memory compute would benefit your dominant kernel.
6	May 6	Wed	Codefest	CMAN (no AI): Energy cost comparison across CPU+DRAM, GPU+HBM, and ideal PIM for a 1024×1024 MVM. CLLM (AI OK): Design and review a 4×4 crossbar MAC unit in SystemVerilog.	Design refinement: complete missing data paths; expand testbench coverage.
7	May 11	Mon	Lecture	Neuromorphic architectures: SpiNNaker, TrueNorth, NorthPole, Loihi; event-driven computation; memristive devices overview.	Assess whether sparsity-aware or neuromorphic changes fit your design.
7	May 13	Wed	Codefest	CMAN (no AI): Sparsity breakeven analysis for CSR vs. dense MVM (N=512). CLLM (AI OK): LLM code review — leaky integrate-and-fire neuron in SystemVerilog; neuromorphic architecture comparison.	Scope assessment: confirm or adjust scope; document decision.
8	May 18	Mon	Lecture	Energy-efficient inference: quantization-aware training vs. post-training quantization; structured vs. unstructured pruning; hardware support for sparse computation; approximate computing.	Apply energy and sparsity analysis to current design.
8	May 20	Wed	Codefest	CMAN (no AI): Architectural specification exercise — 100 TOPS / 5W INT8 inference accelerator design space. CLLM (AI OK): Quantization and pruning sweep on project model; Pareto curve of accuracy vs. speedup.	**M3 DUE Sun, May 24**: Synthesis attempt or justified scope adjustment with full documentation.
9	May 25	Mon	—	Memorial Day (no class).	(no class)

Wk	Date	Day	Type	Topic / activity	Project / milestone
9	May 27	Wed	Codefest	Note: no Monday lecture this week (Memorial Day). CMAN draws on Weeks 2, 3, 8. CMAN (no AI): Arithmetic intensity of your project kernel from first principles; roofline sketch. CLLM (AI OK): Benchmark accelerator vs. SW baseline; plot on roofline; document remaining tasks.	Final benchmarking run; compare accelerator vs. SW baseline from M1.
9			Quiz	Quiz 2: Canvas-based, open-note, individual. Covers lecture material through Week 8.	
10	Jun 1	Mon	Lecture	Emerging technologies: unconventional computing, chiplets (UCIe), post-CMOS paradigms, course synthesis.	Prepare design justification narrative; review AI exam protocol.
10	Jun 3	Wed	Codefest	CMAN (no AI): Closed-book design space retrospective (30 min, from memory). CLLM (AI OK): Final design justification report draft — all 9 sections, ≥1,200 words. Final project deliverables and reflection due.	M4 validation: finalize design justification report; finalize source code, synthesis results, benchmarks. **M4 DUE Sun, Jun 7**: Full deliverable package (source, synthesis, benchmarks, report).
11	Jun 8–13	—	Finals	Final examination (format TBA).	Final exam completed.

Classroom etiquette and expectations:

I'm asking you **to live the "campsite rule"** (inspired by Dan Savage): **leave the place better than you found it. In every aspect. We should miss you if you are not in class!** The campsite rule translates roughly into the items below. The list is incomplete, but you get the idea.

- Listen, engage, participate, contribute.
- Act professionally.
- Respect each other, be kind, show empathy.
- Help each other to be successful.
- Have a growth mindset.
- Have intellectual humility and ambition.
- Up your game, set high standards for yourself.
- Do always a little more instead of just the bare minimum.
- Produce work that you can be proud of.
- Put yourself in the driver's seat, don't blame others.
- Arrive on time, be seated and ready when class starts.
- Do not leave early, do not start to pack up your stuff before class ends.
- No side conversations.
- Do not distract others with your devices, e.g., by playing games, checking social media, texting.
- Remove earphones and sunglasses.
- Do the work you are supposed to do, e.g., don't work on other class assignments. If there are deliverables, submit them at the end of the class session.

Attendance:

"Showing up is 80 percent of life." — Woody Allen

If you want to be successful in this class, you need to show up. Attendance is the easiest way to stay on top of the material, get help, learn from others, and calibrate your work against your peers. And nothing can replace the connections you build in person. Being on time and seated when class starts is a courtesy to

your teacher and classmates, and a sign of professionalism and good work ethic. Please be ready when class begins.

Homework:

For a 4-credit course, you can expect up to 8 hours of homework per week.

Assignments and deliverables:

This course uses specifications grading. Each graded deliverable is evaluated as ***Satisfactory (S)*** or ***Not Yet (NY)*** against clearly defined specifications. Students who receive Not Yet can revise and resubmit. Final course grades are determined by which bundle of satisfactory work a student completes, not by accumulating points.

Codefest deliverables (10 total)

Each Wednesday codefest produces a CMAN+CLLM deliverable pair submitted to your GitHub repository by the following Sunday at 11:59 pm. CMAN must be completed without AI assistance and is submitted as a handwritten photo or plain-text file with all working shown. CLLM scales the same concept to real-world complexity where AI tooling is permitted and required. Each CMAN+CLLM pair is graded *S/NY*. Students receiving *NY* may revise and resubmit once within one week of receiving feedback. Every student will present their work to the class at least once during the term; the instructor will schedule presentations across sessions.

Project milestones (4 total)

The term-long co-design project is assessed at four milestones. Each milestone is graded *S/NY*. Milestones 1 through 3 may be revised once.

- **Milestone 1** (due Sun Apr 13): Refined Heilmeier answers, software profiling results, roofline analysis, HW/SW partition proposal, hardware interface selection with bandwidth justification, initial SW benchmark.
- **Milestone 2** (due Sun May 4): First working simulation with testbench; interface module in HDL with functional testbench; precision choice documented.
- **Milestone 3** (due Sun May 24): Synthesis attempt with timing and area report; interface integrated with compute core; co-simulation demonstrating end-to-end data flow; or justified scope adjustment.
- **Milestone 4** (due Sun Jun 7): Full deliverable package including source code, synthesis results, benchmark comparison, roofline plot, and design justification report. This is final and cannot be revised.

Final examination

During week 10, every student submits their M4 package. An LLM examiner is provided the documentation and creates a set of exam questions on architectural decisions, tradeoffs, and applied concepts. Students are also assessed on their ability to identify when the AI examiner's questions are incorrect or misleading. The final exam format will be announced at a later stage. The final exam is graded *Pass with Distinction* / *Pass* / *No Pass*. No retake is permitted.

Quizzes (2 total)

Two Canvas-based quizzes (Weeks 5 and 9) cover conceptual material from lectures. Open-note, individual, and graded *S/NY* with an 80% threshold.

Revision tokens

Each student begins the term with **3 revision tokens**. A token is spent each time a student revises a Not Yet deliverable. **Tokens cannot be used for Milestone 4 or the final examination**. Additional tokens can be earned in the following ways:

- Spot-check explanations: the instructor conducts 1–3 spot checks with every student during codefests across the term. A clear, correct verbal explanation over all spot checks earns 1 token.
- Codefest presentations: every student presents their work at least once during the term. A strong presentation earns 1 token.
- Consistent Slack contributions throughout the term, as assessed by the instructor, earns 1 token.

Grading:

"*A plumber doesn't get credit for effort; he gets credit if the faucet stops leaking.*" – Seth Godin

Final letter grades are determined by the bundle of *Satisfactory (S)* work a student completes. All requirements in a row must be met. M4 and the final examination cannot be revised; see the penalty rule below.

Penalty for *No Pass* on M4 or final examination

If M4 or the final examination is *No Pass,* the bundle grade drops by **two letter grades** (e.g., A becomes B+). If both M4 and the final examination are No Pass, the grade drops by 4 letter grades (e.g., A becomes B-). No revision is permitted for either deliverable.

Undergraduate grade bundles (ECE 410)

Grade	Codefest deliverables (of 10)	Project milestones (of 4)	Final exam	Quizzes (of 2)
A	8+ S	All 4 S	Pass w/ Distinction	Both S
A−	7+ S	All 4 S	Pass	Both S
B+	6+ S	3+ S	Pass	Both S
B	5+ S	3+ S	Pass	1+ S
B−	4+ S	2+ S	Pass	1+ S
C+	3+ S	2+ S	Pass	0+ S
C	2+ S	1+ S	Pass	0+ S
C−	1+ S	1+ S	Pass	0+ S
D+	0 S	0+ S	Pass	0+ S
D	0 S	0+ S	Pass	0+ S
F	0 S	0 S	No pass	0 S

Graduate and B+M grade bundles (ECE 510)

Grade	Codefest deliverables (of 10)	Project milestones (of 4)	Final exam	Quizzes (of 2)
A	9+ S	All 4 S	Pass w/ Distinction	Both S
A−	8+ S	All 4 S	Pass	Both S
B+	7+ S	3+ S	Pass	Both S
B	6+ S	3+ S	Pass	1+ S
B−	5+ S	2+ S	Pass	1+ S
C+	4+ S	2+ S	Pass	0+ S
C	3+ S	1+ S	Pass	0+ S
C−	2+ S	1+ S	Pass	0+ S
D+	1+ S	0+ S	Pass	0+ S
D	1+ S	0+ S	Pass	0+ S
F	0 S	0 S	No pass	0 S

Make-up assignments and extra credit:

There are no individual make-up assignments or extensions in this class. A late submission is graded Not Yet and consumes one revision token. Late work beyond token availability receives a permanent NY.

Tokens cannot be used for Milestone 4 or the final examination. Additional tokens can be earned through spot-check explanations, codefest presentations, and consistent Slack contributions. See the Assignments and deliverables section for the full list.

How to get help with class-related questions?

1. Post your question(s) on Slack.
2. Come to my office hour: http://www.teuscher-lab.com/officehour
3. Send an e-mail to teuscher@pdx.edu. I always try to reply to student e-mails promptly, but please do not assume that you will get a reply within hours, especially not after 5pm and on weekends. Due to the volume of e-mails I get, my response times can be longer than 24h.

Letters of recommendation:

I am happy to write letters of recommendation for students who engaged and actively participated, students who cared, showed impeccable behavior, were role models for others, and showed leadership skills. Please give me enough advance notice and provide me with your resume and any other relevant information. I cannot usually accommodate last-minute requests unless I have already written a letter for you. How to ask for a recommendation: http://www.chronicle.com/article/How-to-Ask-for-a/235968. If you ask me to write a letter, you will need to complete the following questionnaire: http://bit.ly/CTRecommendationRequest.

Student conduct and plagiarism:

Academic integrity is a vital part of the educational experience at PSU. The Student Code of Conduct is the university's policy on academic dishonesty. A confirmed violation of that code in this course may result in failure of the course.
The use of LLMs is not merely permitted but actively encouraged. In this course, the use of AI tools does not need to be acknowledged. It is understood that you will be using them. However, proper credit must be given for any other source your use.
You must be able to explain any code in your repository. The final examination is designed to verify this. LLM-generated code that you do not understand and cannot debug is a liability, not an asset.

Incomplete grades policy:

Students do not have a right to receive or demand an incomplete grade. The option of assigning an incomplete grade is at the discretion of the instructor when these criteria are met:

- Required satisfactory course completion/participation.
- Reasonable justification for the request.
- Not sought as a substitute for a poor grade.
- Presence of a written agreement (such as the Incomplete Contract).
- A plan in place to resolve Incomplete status.

The Office of the Registrar's website has the full Incomplete Grades policy.

Disability access statement:

If you have, or think you may have, a disability that may affect your work in this class and feel you need accommodations, contact the Disability Resource Center to schedule an appointment and initiate a conversation about reasonable accommodations. The DRC is at 116 Smith Memorial Student Union, 1825 SW Broadway; 503-725-4150; drc@pdx.edu.

Basic needs:

It can be challenging to do your best in class if you have trouble meeting basic needs like safe shelter, sleep, and nutrition. Resource centers across campus are here to provide assistance, referrals, and support. Please contact anyone on this list for assistance:

- Basic Needs Hub: basicneedshub@pdx.edu
- Portland State Food Pantry: psufp.com or pantry@pdx.edu
- C.A.R.E. Program: askdos@pdx.edu, (503) 725-4422
- Student Health & Counseling: askshac@pdx.edu, (503) 725-2800

Understanding sexual misconduct:

PSU desires to create a safe campus. As part of that mission, PSU requires all students to take the Understanding Sexual Misconduct and Resources learning module. If you or someone you know has been harassed or assaulted, you can find the appropriate resources on PSU's Sexual Misconduct Response website.

Title IX reporting:

Students frequently come to me for assistance in matters not related to the course material. Please be aware that PSU's policies require instructors to report any instance of sexual harassment, sexual and relationship violence and/or other forms of prohibited discrimination to university officials, who will keep the information private. If you would rather share information about these matters with a PSU staff member who does not have these reporting responsibilities and can keep the information confidential, please use these campus resources:

- Confidential Advocates: 503-894-7982 or schedule online (for matters regarding sexual harassment and sexual and relationship violence)
- Center for Student Health and Counseling: 1880 SW 6th Avenue #200; 503-725-2800

Discrimination and bias incidents:

The Office of Equity and Compliance (OEC) addresses complaints of discrimination, discriminatory Harassment, and sexual harassment against employees (faculty and staff). If you or someone you know believes they have been discriminated against, you may file a complaint. Someone from the OEC will contact you to discuss how to best address your complaint.
The Bias Review Team (BRT) gathers information on bias incidents that happen on and around campus, and gives resources and support to individuals who experience them. You can report a bias incident you experienced or learned about. A member of the BRT will contact you if you indicate you would like to be contacted.

Cultural resource centers:

Cultural Resource Centers (CRCs) create a student-centered inclusive environment that enriches the university experience. They honor diversity, explore social justice issues, celebrate cultural traditions, and foster student identities, success, and leadership. They provide opportunities for student leadership, employment, and volunteering; student resources such as computer labs, event, lounge, and study spaces; and extensive programming. All are welcome!

- Multicultural Student Center
- La Casa Latina Student Center
- Native American Student and Community Center
- Pan African Commons
- Pacific Islander, Asian, and Asian American Student Center
- Middle East, North Africa, South Asia Initiative

Recording technology notice:

We may use technology for virtual meetings and recordings in this course. Our use of such technology is governed by FERPA, the Acceptable Use Policy and PSU's Student Code of Conduct. A record of all meetings and recordings is kept and stored by PSU, in accordance with the Acceptable Use Policy and FERPA. I will not share recordings of your class activities outside of course participants, which include your fellow students, TAs/GAs/Mentors, and any guest faculty or community-based learning partners that we may engage with. **You may not share recordings outside this course. Doing so may result in disciplinary action.**

In an emergency, call:

Immediate Threat to Life and Safety *For issues such as: a medical emergency, urgent violent incident, fire, etc.*	**503-725-5911** **or 9-1-1**
Campus Public Safety Office (CPSO) – Non-Emergency *For non-emergency issues such as: vandalism, disturbance, suspicious person, theft, suspicious packages, access control, etc.*	**503-725-4407**
PSU CARES *Connects students with resources when they are experiencing distress or other concerns that might impact their personal or academic success.*	**MAKE A REFERRAL** **503-725-4422** askdos@pdx.edu